Team DNA Hackers

Project Phase 1

Data Requirements

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1 - Mini World Development [50 Marks]

1.1 Idea Formulation [20]

- **What is our Fantasy mini world**
  Our mini world is inspired by the Carmen Sandiego universe, a detective and adventure series that revolves around high-profile thefts and international investigations.
  The story follows Carmen Sandiego, a master thief who travels across the globe carrying out complex heists — sometimes for noble reasons. Opposing her is ACME Detective Agency, a global network of agents tracking her and her team of thieves through clues, artifacts, and travel patterns.
- **Our proposed database system + How it integrates with the world**
  We propose to build AID: ACME INTELLIGENCE DATABASE which is a Global Crime Tracking and Investigation Database System used by the ACME Detective Agency. ACME is an international detective agency that organizes missions to recover stolen artifacts and catch villains.
  Our database manages and analyzes all the intelligence data collected from missions around the world — including suspects, criminal profiles, agent assignments, stolen artifacts, locations, and the complex relationships between them. It helps detectives track the movement of criminals, connect clues to crimes, and generate intelligence reports that predict future thefts.
  This structure enables cross-entity analysis, intelligence reporting, and operational insights, which are crucial for preventing and solving crimes on a global scale.
- **How it integrates with the world**
  Detectives use the database to record and retrieve information about crimes and suspects.
  It stores clues and evidence collected during missions, linking them to specific cases and locations.

It helps in identifying patterns — for example, if the same suspect or artifact type appears in multiple heists.
The system can analyze travel data to predict where the next heist might occur.
Reports from the database guide agents during missions and help in resource allocation

- **Novelty**
  Our proposed database is unique because it combines storytelling, intelligence analysis, and predictive crime tracking in a single interconnected system.

  Key aspects:

  - Predictive Intelligence: Uses historical mission data to forecast future thefts.
  - Global Data Interconnectivity: Links missions, suspects, and clues across geography, artifacts, and criminal profiles.
  - Gamified Interface: Interactive mission trails enhance engagement and training.
  - Hybrid Ethical Analysis: Distinguishes between noble and malicious crimes.
  - Supports Advanced Analytics: Enables derived attributes and complex reports, such as top villains, most targeted countries, and agent performance metrics.

1.2 User Identification & Interaction [15 Marks]

In our Carmen Sandiego – Global Crime Tracking and Investigation System, the primary **users** of the database are **ACME detectives, intelligence analysts, and administrative officers** who operate the system to monitor global criminal activity.

- **ACME Detectives** use the database to record new cases, update mission progress, and retrieve intelligence such as known suspects, stolen artifacts, or crime locations. They can also add new clues or evidence discovered during field operations.
- **Intelligence Analysts** interact with the database to generate analytical reports and identify hidden patterns — such as connections between villains, repeating locations, or similar types of stolen artifacts. They primarily use retrieval, aggregate, and search operations to derive insights.

- **Administrative Officers** manage user accounts, validate data integrity, and authorize information sharing between departments. They perform modification operations like adding or removing agents, updating case status, and maintaining referential integrity.

1.3 Purpose of the Database [15 Marks]

State clearly why the database is important in the mini-world setting. What does it offer that non-DB solutions don't?

The Global Crime Tracking and Investigation **Database is essential** in the *Carmen Sandiego* universe as

- It acts as a centralized system for managing all data related to global crimes, missions, suspects, and clues.
- It helps ACME agents collect, organize, and analyze intelligence efficiently across different countries.

Unlike non-database systems such as manual records, spreadsheets, or isolated files, this database provides real-time data sharing, cross-case analysis, and relationship tracking between multiple entities (e.g., agents, villains, and locations).

A non-database system cannot handle large, interconnected data, maintain data integrity automatically, or perform complex queries such as finding links between multiple crimes or predicting future heists based on patterns.

The database ensures data accuracy, consistency, and security, and allows automatic constraint checking and advanced search functions. It transforms scattered information into a unified, intelligent investigation platform, making the entire crime-solving process faster, more coordinated, and data-driven.

Data Integrity & Consistency: Enforces rules (e.g., an agent assigned to a mission must exist in the Agents table) through constraints and foreign keys, which is impossible with flat files.

Complex Querying: The ability to perform multi-table JOIN operations is critical for generating intelligence. An analyst must be able to ask, "Show me all agents who have encountered V.I.L.E. operatives in Southeast Asia and specialize in forensics."

Concurrency: Allows multiple users (agents, analysts) to access and modify data simultaneously without corruption.

Scalability & Performance: Can efficiently handle millions of records related to missions, artifacts, and evidence logs, which would be unmanageable in spreadsheets.

Security & Access Control: Provides robust, role-based security, ensuring field agents cannot access sensitive analytical reports and analysts cannot alter raw evidence logs.

2 - Database Requirements [50 Marks]

Your mini-world should have at least:

- six entity types
- two entities with at least two-key attributes
- two weak entity types
- seven relationship types (this should include cardinality ratios and participation constraints)
- two n > 3 relationship types
- at least one sub class hierarchy
- few composite, multi-valued and derived attributes

Sidenote: Specify constraints that your data may have. Specify the acceptable values, ranges, and formats for the attributes of the data. Clearly define the cardinality and participation constraints of the relationships. Make sure that there are integrity constraints and enforcement mechanisms for them.

• **six entity types** (we have 10 as shown below)

1. Agent
   - Primary Key: AgentID (INT)
   - Attributes:
     - FirstName, LastName (VARCHAR)
     - Rank (ENUM: 'Junior','Field','Senior','Lead','Analyst')
     - Specialties (MULTI-VALUED: e.g., {'Forensics','Linguistics'})
     - Contact (COMPOSITE: Phone, Email)
     - Country (VARCHAR)
     - HireDate (DATE)
     - ExperienceYears (DERIVED: YEAR(CURDATE()) - YEAR(HireDate))

   Purpose: Stores detailed profiles of ACME agents and their professional information.

2. Mission

- Primary Key: MissionID (INT)
- Attributes:
  - MissionName (VARCHAR)
  - Objective (TEXT)
  - StartDate (DATE)
  - EndDate (DATE, NULL)
  - Status (ENUM: 'Planned','Active','Completed','Failed','Aborted')
  - Region (VARCHAR)
  - DurationDays (DERIVED: DATEDIFF(EndDate, StartDate))

Purpose: Tracks each mission's details, progress, and completion statistics.

3. Criminal

- Primary Key: CriminalID (INT)
- Attributes:
  - Name, Alias (VARCHAR)
  - Nationality (VARCHAR)
  - SkillSet (MULTI-VALUED)
  - Status (ENUM: 'AtLarge','Apprehended','Deceased')
- Subclass Hierarchy: Leader and Operative

Purpose: Maintains detailed information on criminals, their skills, and hierarchical structure.

4. Organization

- Primary Key: OrgID (INT)
- Attributes:
  - OrgName (VARCHAR)
  - Type (ENUM: 'CriminalGroup','Agency','Sponsor')
  - HQ_LocationID (FK → Location.LocationID)

Purpose: Stores data about criminal and law enforcement organizations such as V.I.L.E. and ACME.

5. Artifact

- Primary Key: ArtifactID (INT)
- Attributes:
  - ArtifactName (VARCHAR)
  - OriginCountry (VARCHAR)
  - HistoricalValue (DECIMAL)
  - CurrentStatus (ENUM: 'Secure','Missing','InTransit','OnDisplay')
  - PreviousLocations (MULTI-VALUED)

Purpose: Tracks historical artifacts that are often targeted or recovered during missions.

6. Location

- Primary Key: LocationID (INT)
- Attributes:
  - City, Country, Region (VARCHAR)
  - Latitude, Longitude (DECIMAL)
  - RiskLevel (INT: 1–10)

Purpose: Represents geographical data for missions, organizations, and artifact tracking.

7. AgentAssignment (Two-Key Entity)

- Composite Primary Key: (AgentID, MissionID)
- Attributes: Role (ENUM: 'Lead','Support','Observer','Forensics'), AssignedOn (DATE)

Purpose: Links agents to missions and defines their specific roles.

8. ArtifactMovement (Two-Key Entity)

- Composite Primary Key: (ArtifactID, MovementNo) (*MovementNo acts as a sequential number for each artifact's movements*)
- Attributes:
  - ArtifactID → *Artifact.ArtifactID* (Foreign Key)
  - FromLocationID → *Location.LocationID* (Foreign Key)
  - ToLocationID → *Location.LocationID* (Foreign Key)
  - MovedOn (DATETIME, NOT NULL) — *Date and time of transfer.*
  - MoveReason (VARCHAR, CHECK (MoveReason IN ('Recovered','Exhibit Transfer','Security Risk','Smuggling Attempt')))
  - HandledBy_AgentID (FK → Agent.AgentID, NULLABLE) — *Agent responsible for the transfer.* (Foreign Key)

Purpose: Tracks how artifacts are transferred between locations, the reason for each movement, and the responsible agents. This entity ensures historical tracking of artifact logistics across missions and prevents data duplication.

9. Evidence (Weak Entity)

- Identifying Relationship: Mission → Evidence
- Partial Key: EvidenceNo (INT)
- Identifying Owner's Key: MissionID (FK → Mission.MissionID)
- Composite Primary Key: (MissionID, EvidenceNo)
- Other Attributes:
  - Type (ENUM: 'Photo','DNA','Fingerprint','Document','Video')
  - CollectedBy_AgentID (FK → Agent.AgentID)
  - ChainOfCustody (TEXT)
  - Status (ENUM: 'Unprocessed','InLab','Verified','Contaminated')

Purpose: Represents evidence items collected during missions. Each piece of evidence exists only in the context of a specific mission (hence a weak entity).

10. MissionTask (Weak Entity)

- Identifying Relationship: Mission → MissionTask
- Partial Key: TaskNo (INT)
- Identifying Owner's Key: MissionID (FK → Mission.MissionID)
- Composite Primary Key: (MissionID, TaskNo)
- Other Attributes:
  - Description (TEXT)
  - AssignedAgentID (FK → Agent.AgentID)
  - Status (ENUM: 'Pending','Ongoing','Completed')

Purpose: Represents tasks or sub-missions that belong to a specific mission. Each task cannot exist independently without its parent mission.

11. CriminalInvolvement (Two-Key Entity)

- Composite Primary Key: (CriminalID, MissionID)
- Attributes:
  - CriminalID → *Criminal.CriminalID (*Foreign Key*)*
  - MissionID → *Mission.MissionID (*Foreign Key*)*
  - Role (ENUM: 'Planner','Field Operative','Mastermind') — *Defines the part played by the criminal in the mission.*
  - Arrested (BOOLEAN, DEFAULT FALSE) — *Indicates whether the criminal was apprehended during or after the mission.*
  - Reward (DECIMAL(10,2), CHECK (Reward >= 0)) — *Specifies the monetary reward associated with capturing the criminal for that mission.*

Purpose: Establishes a many-to-many relationship between *Criminal* and *Mission* entities. It records each criminal's involvement in specific missions, their operational role, and whether they were captured, enabling efficient tracking of criminal activity and outcomes across missions.

• two entities with at least two-key attributes

ArtifactMovement, AgentAssignment, CriminalInvolvement

• two weak entity types

- Evidence → Weakly dependent on *Mission*.
- MissionTask → Weakly dependent on *Mission*.

• seven relationship types (this should include cardinality ratios and participation constraints)

1. Agent ASSIGNED_TO Mission (M:N) - Implemented via AgentAssignment.
   - Agent (0,N): An agent can be on zero or many missions. (Partial participation)
   - Mission (1,N): A mission must have at least one agent. (Total participation)
2. Mission TARGETS Artifact (M:N) - Implemented via a MissionArtifacts linking table.
   - Mission (1,N): A mission must target at least one artifact. (Total participation)
   - Artifact (0,N): An artifact may be the target of zero or many missions over time. (Partial participation)
3. Criminal MEMBER_OF Organization (N:1)
   - Criminal (1,1): Each criminal belongs to exactly one main organization. (Total participation)

- Organization (0,N): An organization can have zero or many members. (Partial participation)

4. Mission COLLECTS Evidence (1:N)
   - Mission (1, 1..N): A piece of evidence must belong to exactly one mission. (Total participation for Evidence)
   - This is an identifying relationship for the weak entity Evidence.
5. Mission COMPRISES MissionTask (1:N)
   - Mission (1, 1..N): A task must belong to exactly one mission. (Total participation for MissionTask)
   - This is an identifying relationship for the weak entity MissionTask.
6. Agent TRACKS Criminal (M:N) - Implemented via an AgentTracking linking table.
   - Agent (0,N): An agent may track many criminals.
   - Criminal (0,N): A criminal may be tracked by many agents.
7. Artifact MOVED_BETWEEN Locations - Implemented via the ArtifactMovement entity.
   - A single movement record links one artifact to two locations (From and To).

• **two n > 3 relationship types**

Quaternary Relationship (n=4): OperationLog

- Participating Entities: Mission, Agent, Artifact, Location.
- Semantics: Records a specific action performed by an Agent on an Artifact at a specific Location during a Mission.
- Implementation: A table OperationLog with foreign keys: MissionID, AgentID, ArtifactID, LocationID, plus other attributes like ActionType and Timestamp.

Ternary Relationship (n=3): IntelShare

- Participating Entities: Agent (sender), Agent (receiver), Criminal (subject).
- Semantics: Logs an intel sharing event where one agent sends information about a specific criminal to another agent.
- Implementation: A table IntelShare with foreign keys: SenderAgentID, ReceiverAgentID, CriminalID, plus a ShareDate attribute.

• **at least one sub class hierarchy**

Superclass: Criminal

- Subclass: Leader (Attributes: LeadershipTier, CommandRegion)

- Subclass: Operative (Attributes: Specialty, HandlerID (FK to a Leader's CriminalID))

• **few composite, multi-valued and derived attributes**

Composite: Agent.Contact can be broken down into (Phone, Email).
Multi-valued: Agent.Specialties can hold a list of skills like {'Forensics', 'Linguistics'}.
Derived: Agent.ExperienceYears is not stored in the database. It's calculated in real-time: CURRENT_DATE - HireDate.

• **Global Constraints & Validations**

- Every Mission must have at least one assigned Agent.
- An Artifact cannot be “InTransit” without a corresponding ArtifactMovement entry.
- A Criminal marked “Apprehended” must appear in at least one CriminalInvolvement record with Arrested = TRUE.
- Locations must have RiskLevel between 1 and 10.
- Agent HireDate ≤ CURRENT_DATE.

3 - Functional Requirements [50 Marks]

In your mini-world, you will be required to have applications that operate on the database.

The **Carmen Sandiego Global Crime Tracking Database** supports multiple operations that enable agents and analysts to **retrieve, analyze, and modify data** efficiently.
These operations assist ACME in monitoring missions, tracking criminals, analyzing movements, and maintaining database integrity across multiple entities.

3.1 Retrieval Operations [20 Marks]

At least one query function for each should be described. It need not be in the SQL syntax but can be explained in English on what the function/query intends to accomplish.

3.1.1 Selection

Example: ”Retrieve all premium customers who joined in 2025”

Query: Retrieve all *Active Missions* currently taking place in *Europe* with a *Risk Level ≥ 7* at their associated locations.
Purpose: Helps identify high-risk missions that require urgent attention or additional agent support.

3.1.2 Projection

Example: ”Display names and contact details of all active customers”

Query: Display *Agent Names*, *Contact Details*, and *Specialties* of all *Senior* and *Lead* agents.
Purpose: Provides an overview of available expert agents for task allocation or mission reassignment.

3.1.3 Aggregate (SUM, MAX, MIN, AVG, etc.)

Example: ”Calculate total revenue generated this quarter”

Query: Calculate the *Average Mission Duration (in days)* for all *Completed Missions*.
Purpose: Enables ACME analysts to measure overall operational efficiency and identify delays in fieldwork.

3.1.4 Search

Example: Partial text matching for entries in an entity, ”TECH” will match with ”TECHNOL-
OGY” or ”FINTECH”

Query: Search for all *Artifacts* whose names contain the keyword “Crown” (e.g., “Crown Jewel”, “Emerald Crown”).
Purpose: Allows quick identification of specific stolen or recovered artifacts through partial text matching.

3.2 Analysis Report (Minimum 3) [15 Marks]

Generate comprehensive intelligence reports that demonstrate relationships between entities and provide actionable insights. These reports should use Join operations across multiple entities, not simple selections from single entities.
Example: ”Monthly revenue breakdown by customer segment and service type”

These reports are generated using **join operations across multiple entities**, **Report 1: Mission Performance Summary**

Description: For each *Mission*, display the *Lead Agent*, *Total Evidence Collected*, and *Mission Status*.
Entities Used: Mission, AgentAssignment, Agent, Evidence
Purpose: Helps management evaluate which agents and missions yield the highest evidence of success rates.

Report 2: Criminal Network Mapping

Description: Generate a report listing all *Criminals* linked to a specific *Organization* (e.g., V.I.L.E.), including their *Roles (Leader/Operative)* and *Current Status*.
Entities Used: Criminal, Organization
Purpose: Visualizes the criminal hierarchy, assisting in dismantling networks by identifying key leaders.

Report 3: Artifact Movement Analysis

Description: Show all *Artifacts* that have been moved more than twice in the past month, along with their *FromLocation*, *ToLocation*, and *MoveReason*.

Entities Used: Artifact, ArtifactMovement, Location

Purpose: Detects suspicious movement patterns to prevent smuggling or theft of valuable artifacts.

3.3 Modification Operations [15 Marks]

(a) At least one insertion of data with automatic checking for violations of integrity constraints

Scenario: Add a new *Mission* record along with initial *AgentAssignments* and *Tasks*.

Automatic Integrity Checks:

- Validates that assigned *Agents* exist in the database.
- Ensures *StartDate* < *EndDate*.
- Sets *Status* = *'Planned'* by default.

(b) At least one update operation with enforcement of cascading business rules

Scenario: When a *Mission* status is updated to *"Completed"*, all related *MissionTasks* are automatically set to *"Completed"*, and *DurationDays* is recalculated.

Cascading Rule: Keeps mission and task status synchronized to maintain data consistency.

(c) At least one deletion operation maintaining referential integrity

Scenario: When a *Mission* is deleted, all associated *Evidence* and *MissionTasks* are automatically deleted (cascade delete), but *Agent* and *Artifact* records remain intact.

Referential Integrity: Ensures dependent data is removed properly, avoiding orphaned records.